CSA1708-ARTIFICIAL INTELLIGENCE

LAB EXPERIMENTS

Program 12: WRITE THE PYTHON POGRAM FOOR TIC TAC TOE GAME

AIM

To write a Python program to implement the Tic-Tac-Toe game for two players.

ALGORITHM

1. Start the program.
2. Create a 3 × 3 game board initialized with empty spaces.
3. Display the game board.
4. Ask the current player to enter a position between 1 and 9.
5. Check whether the selected position is valid and empty.
6. Place the current player's symbol (X or O) in the selected position.
7. Check whether the current player has won by completing a row, column, or diagonal.
8. If a player wins, display the winner and stop the game.
9. If the board is full and nobody wins, declare a draw.
10. Otherwise, switch players and repeat the process.
11. Stop the program.

untitled (3.14.8)

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```
board = [' '] * 9

def display():
    for i in range(0, 9, 3):
        print(board[i], '|', board[i+1], '|', board[i+2])
        if i < 6:
            print('--+---+--')

player = 'X'

for turn in range(9):
    display()
    pos = int(input(f"Player {player}, enter position (1-9): ")) - 1

    if pos < 0 or pos > 8 or board[pos] != ' ':
        print("Invalid position")
        continue

    board[pos] = player

    wins = [(0,1,2), (3,4,5), (6,7,8),
            (0,3,6), (1,4,7), (2,5,8),
            (0,4,8), (2,4,6)]

    if any(all(board[i] == player for i in line) for line in wins):
        display()
        print(player, "wins!")
        break

    player = 'O' if player == 'X' else 'X'
else:
    display()
    print("Draw!")
```

IDLE Shell 3.14.8

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```
Python 3.14.8 (tags/v3.14.8:8e6e75d, Sep 30 2026, 18:19:33) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/praji/AppData/Local/Programs/Python/Python314/python 12.py =
  |   |
--+---+--
  |   |
--+---+--
  |   |
Player X, enter position (1-9):
```

RESULT

The Python program was executed successfully, implementing the Tic-Tac-Toe game for two players. The game checks for a win, a draw, and invalid moves.